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REMOVAL TOOL FOR CORE OF AIR VALVE

Abstract

A removal tool is applied to remove a Presta valve, the Presta valve includes a Presta core, the Presta core includes a fixing portion with screw threads, and the fixing portion of the Presta valve has two drive faces not having screw threads. The fixing portion of the Presta valve also has a valve head. The removal tool includes a body. The body includes a Presta detachable head which has a Presta cutout, two Presta actuation planes, and a through orifice. A Presta flexible positioning means is fixed on the Presta cutout of the Presta detachable head of the body and is staggered from the two Presta actuation planes, and the Presta flexible positioning means corresponds to the fixing portion of the Presta core.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a removal tool, and more particularly to the removal tool for a core of an air valve.

BACKGROUND

[0002] Conventionally, when filling tire sealant into a tire from an air valve, a core of the air valve is removed. However, the core will be broken since the air valve is damaged after a period of use time, thus causing an unstable air pressure or an air leakage. Therefore, regular inspection and replacement of the core will help ensure that the tire is properly inflated, thus improving a driving safety. When the core inside the air valve of the tire needs to be replaced or routinely maintained, the core must be removed as well.

[0003] Referring to FIGS. 14-16, a conventional removal tool **80** for a core of an air valve contains a grip portion **81** and a detachable head **82**, wherein the grip portion **81** is configured to be grasped by a user, and the detachable head **82** has an opening **821** defined on an edge thereof, and the opening **821** has a receiving orifice **822** formed on a bottom thereof. In use, the receiving orifice **822** is fitted with the core **92** of the air valve **90**, and the opening **821** is engaged with the core **92** of the air **90** so as to rotate the core **92** away from the air valve **90**.

[0004] The core **92** is detachable easily by using the conventional removal tool **80** but cannot be fixed, so when removing the core **92** from the air valve **90**, the core **92** cannot be removed from the air valve **90** by using the removal tool **90** directly, thus manually removing the core **92** from the air valve **90** is required. When removing the core **92** from the air valve **90**, the core **92** cannot be fixed by using the removal tool **80** to cause a dropping, thus having a troublesome removal.

[0005] The present invention has arisen to mitigate and/or obviate the afore-described disadvantages.

SUMMARY

[0006] The primary aspect of the present invention is to provide a removal tool for a core of an air valve which is simplified to reduce related components, be connected easily, accelerate a connecting process, obtain a stability, lower a fabrication cost, prolong a service life, avoid a removal of the cores, have an easy operation, and enhance a using practicality.

[0007] To obtain the above-mentioned aspect, a removal tool for a core of an air valve provided by the present invention contains a body, and the body includes a Presta detachable head formed on an end thereof. The Presta detachable head has a Presta cutout formed on an edge thereof, two Presta actuation planes formed on two sides of the Presta cutout and corresponding to two drive faces of a Presta valve, and the two drive faces have no screw threads. Also, a through orifice is defined on a bottom of the Presta cutout and extends to a top of the Presta cutout.

[0008] The removal tool also contains a Presta flexible positioning means fixed on the Presta detachable head of the body and is staggered from the two Presta actuation planes. The through orifice of the body is fitted with the Presta core, the two Presta actuation planes of the Presta cutout are engaged with the two drive faces, and the Presta flexible positioning means abuts against the Presta core.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a perspective view showing the assembly of a removal tool for a core of an air valve according to a preferred embodiment of the present invention.

[0010] FIG. 2 is a perspective view showing the exploded components of the removal tool

according to the preferred embodiment of the present invention.

[0011] FIG. **3** is a cross-sectional perspective view showing the exploded components of a part of the removal tool according to the preferred embodiment of the present invention.

[0012] FIG. **4** is a side plan view showing the assembly of the removal tool according to the preferred embodiment of the present invention.

[0013] FIG. **5** is a cross-sectional view taken along the line A-A of FIG. **4**.

[0014] FIG. **6** is a cross-sectional view taken along the line B-B of FIG. **4**.

[0015] FIG. **7** is a cross-sectional view taken along the line C-C of FIG. **5**.

[0016] FIG. **8** is a cross-sectional view showing a Presta valve being removed by using the removal tool according to the preferred embodiment of the present invention.

[0017] FIG. **9** is another cross-sectional view showing the Presta valve being removed by using the removal tool according to the preferred embodiment of the present invention.

[0018] FIG. **10** is also another cross-sectional view showing the Presta valve being removed by using the removal tool according to the preferred embodiment of the present invention.

[0019] FIG. **11** is a cross-sectional view showing a Schrader valve being removed by using the removal tool according to the preferred embodiment of the present invention.

[0020] FIG. **12** is another cross-sectional view showing the Schrader valve being removed by using the removal tool according to the preferred embodiment of the present invention.

[0021] FIG. **13** is also another cross-sectional view showing the Schrader valve being removed by using the removal tool according to the preferred embodiment of the present invention.

[0022] FIG. **14** is a perspective view of a conventional removal tool for a core of an air valve.

[0023] FIG. **15** is a cross-sectional view showing the operation of the conventional removal tool for the core of the air valve.

[0024] FIG. **16** is another cross-sectional view showing the operation of the conventional removal tool for the core of the air valve.

DETAILED DESCRIPTION

[0025] With reference to FIGS. **1-7**, a removal tool for a core of an air valve according to a preferred embodiment of the present invention is applied to remove the core **12** of the air valve **11** of a tire **10** (as shown in FIG. **8**), the air valve **11** is a Schrader valve **11A** (as shown in FIG. **8**) or a Presta valve **11B** (as illustrated in FIG. **11**), wherein the Schrader valve **11A** includes a Schrader core **12A** accommodated therein, and the Schrader core **12A** of the Schrader valve **11A** has a fixing portion **121A**, the fixing portion **121A** of the Schrader valve **11A** has two drive faces **122A** formed on two sides thereof and has a push needle **123A** extending from the fixing portion **121A**. The Presta valve **11B** includes a Presta core **12B** accommodated therein, wherein the Presta core **12B** includes a fixing portion **121B** with screw threads, and the fixing portion **121B** of the Presta valve **11B** has two drive faces **122B** formed on two sides thereof and not having screw threads, the fixing portion **121B** of the Presta valve **11B** also has a valve head **123B** extending therefrom. The removal tool includes a body **20**, a Presta flexible positioning means **30**, and a Schrader flexible positioning means **30A**.

[0026] The body **20** includes a Presta detachable head **21** formed on an end thereof, a Schrader detachable head **22** formed on the other end of the body **20**, and a grip portion **23** located away from the Presta detachable head **21** and the Schrader detachable head **22**. The Presta detachable head **21** has a Presta cutout **211** formed on an edge thereof, two Presta actuation planes **212** formed on two sides of the Presta cutout **211**, a through orifice **213** defined on a bottom of the Presta cutout **211** and extending to a top of the Presta cutout **211**, a Presta recess **214** formed on the Presta detachable head **21**, and at least one Presta passing orifice **215** defined in the Presta recess **214** and communicated with the Presta cutout **211**. The two Presta actuation planes **212** are staggered from the at least one Presta passing orifice **215** in a cross shape. In this embodiment, two Presta passing orifices **215** are provided. The Schrader detachable head **22** has a Schrader cutout **221** formed on an edge thereof, two Schrader actuation planes **222** formed on two sides of the Schrader cutout **221**,

a Schrader recess **223** formed on the Schrader detachable head **21**, and at least one Schrader passing orifice **224** defined in the Schrader recess **223** and communicated with the Schrader cutout **221**. In this embodiment, two Schrader passing orifices **224** are provided.

[0027] The Presta flexible positioning means **30** is fixed on the Presta detachable head **21** of the body **20**, and the Schrader flexible positioning means **30A** is fixed on the Schrader detachable head **22**. The two Presta actuation planes **212** are staggered from the Presta flexible positioning means **30** in a cross shape. The Presta flexible positioning means **30** corresponds to the fixing portion **121B** with the screw threads, wherein the Presta flexible positioning means **30** is a Presta clamp element **31**, the Presta clamp element **31** has a flexible Presta connector **32** mounted on an outer wall thereof, and the flexible Presta connector **32** is engaged in the Presta recess **214** and has at least one Presta retainer **33** extending to an inner wall of the flexible Presta connector **32**. In this embodiment, two Presta retainers **33** are provided and are connected with the two Presta passing orifices **215**, and an extending length of the two Presta retainers **33** are greater than a size of the two Presta passing orifices **215** so that a part of the two Presta retainers **33** extend out of the Presta cutout **211**, wherein the part of the two Presta retainers **33** have a Presta arcuate groove **34** extending therefrom, the two Presta retainers **33** of the Presta flexible positioning means **30** of the Presta cutout **211** are staggered from two Presta actuation planes **212** in a cross shape, and the Presta flexible positioning means **30** corresponds to the fixing portion **121B** of the Presta valve **11B**.

[0028] The Schrader flexible positioning means **30A** corresponds to the fixing portion **121A** with screw threads, wherein the Schrader flexible positioning means **30A** is a Schrader clamp element **31A**, the Schrader clamp element **31A** has a flexible Schrader connector **32A** mounted on an outer wall thereof, and the flexible Schrader connector **32A** is engaged in the Schrader recess **223** and has at least one Schrader retainer **33A** extending to an inner wall of the flexible Schrader connector **33A**. In this embodiment, two Schrader retainers **33A** are provided and are connected with the two Schrader passing orifices **224**, and an extending length of the two Schrader retainers **33A** are greater than a size of the two Schrader passing orifices **224** so that a part of the two Schrader retainer **33A** extends out of the Schrader cutout **221**, wherein the part of the two Schrader retainer **33A** has a Schrader arcuate groove **34A** extending therefrom, and the Schrader flexible positioning means **30A** corresponds to the push needle **123A**.

[0029] Referring to FIGS. **8-10**, when desiring to remove the Presta valve **11B**, the Presta detachable head **21** of the body **20** is fitted on the Presta valve **11B** so that the Presta core **12B** extends into the through orifice **213** of the body **20**, the fixing portion **121B** of the Presta valve **11B** is located in the Presta cutout **211**, and the two Presta actuation planes **212** are engaged with the two drive faces **122B** of the fixing portion **121B** of the Presta valve **11B**, wherein the two Presta retainers **33** of the Presta flexible positioning means **30** are flexibly engaged with the fixing portion **121B** of the Presta valve **11B**, then the grip portion **23** of the body **20** is driven to rotate so that the two Presta actuation planes **212** actuate the Presta core **12B** to rotatably remove from the Presta valve **11B**. Since the two Presta retainers **33** of the Presta flexible positioning means **30** are flexibly engaged with the fixing portion **121B** of the Presta valve **11B**, the Presta core **12B** is rotatably removed from the Presta valve **11B** with the body **20**, thus removing the Presta valve **11B** securely.

[0030] As illustrated in FIGS. **11-13**, when desiring to remove the Schrader valve **11A**, the Schrader detachable head **22** of the body **20** is fitted on the Schrader valve **11A** so that the two Schrader actuation planes **122A** are engaged with the Schrader cutout **221**, wherein the two Schrader retainers **33A** of the Schrader flexible positioning means **30A** are flexibly engaged with the push needle **123A**, then the body **20** is rotated to actuate the Schrader core **12A** to remove from the Schrader valve **11A**, and the Schrader valve **11A** is detachable.

[0031] Thereby, the removal tool of the present invention has the following advantages:

[0032] The removal tool is simplified to reduce related components, be connected easily, accelerate a connecting process, obtain a stability, lower a fabrication cost, prolong a service life, avoid a

removal of the cores **12**, have an easy operation, and enhance a using practicality.
[0033] While the first embodiments of the invention have been set forth for the purpose of disclosure, modifications of the disclosed embodiments of the invention as well as other embodiments thereof may occur to those skilled in the art. The scope of the claims should not be limited by the first embodiments set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole.

Claims

1. A removal tool being applied to remove a Presta valve, the Presta valve including a Presta core accommodated in the Presta valve, wherein the Presta core includes a fixing portion with screw threads, and the fixing portion of the Presta valve has two drive faces formed on two sides thereof and not having screw threads, the fixing portion of the Presta valve also has a valve head extending therefrom; wherein the removal tool includes: a body which includes a Presta detachable head formed on an end of the body, wherein the Presta detachable head has a Presta cutout formed on an edge thereof, two Presta actuation planes formed on two sides of the Presta cutout and corresponding to the two drive faces of the Presta valve, and a through orifice defined on a bottom of the Presta cutout and extending to a top of the Presta cutout; and wherein a Presta flexible positioning means is fixed on the Presta cutout of the Presta detachable head of the body and is staggered from the two Presta actuation planes, and the Presta flexible positioning means corresponds to the fixing portion of the Presta core; wherein the Presta core is fitted into the through orifice of the body, the two Presta actuation planes are engaged with the two drive faces of the Presta cutout, and the Presta flexible positioning means abuts against the fixing portion of the Presta valve so that the removal tool actuates the Presta core to rotate away from the Presta valve, thus removing the Presta core.
2. The removal tool as claimed in claim 1, wherein the Presta flexible positioning means is fixed on the Presta cutout and is staggered from the two Presta actuation planes in a cross shape.
3. The removal tool as claimed in claim 2, wherein the Presta detachable head has a Presta recess formed thereon, and at least one Presta passing orifice defined in the Presta recess and communicated with the Presta cutout, wherein the two Presta actuation planes are staggered from the at least one Presta passing orifice in a cross shape; wherein the Presta flexible positioning means is a Presta clamp element, and the Presta clamp element has a flexible Presta connector engaged in the Presta recess, wherein the flexible Presta connector has at least one Presta retainer extending to an inner wall thereof, and the at least one Presta retainer is connected with the at least one Presta passing orifice, wherein an extending length of the at least one Presta retainer is greater than a size of the at least one Presta passing orifice so that the at least one Presta retainer extends out of the Presta cutout.
4. The removal tool as claimed in claim 3, wherein the part of the at least one Presta retainer has a Presta arcuate groove extending therefrom.
5. The removal tool as claimed in claim 1, wherein the body includes a Schrader detachable head formed on the other end thereof and configured to remove a Schrader core of a Schrader valve, the Schrader detachable head includes a Schrader recess formed thereon, at least one Schrader passing orifice defined in the Schrader recess and communicated with the Schrader cutout, and a Schrader flexible positioning means is fixed on the Schrader detachable head, wherein the Schrader flexible positioning means corresponds to the fixing portion of Schrader core and the fixing portion of the Schrader core has screw threads, wherein the Schrader flexible positioning means is a Schrader clamp element, the Schrader clamp element has a flexible Schrader connector mounted on an outer wall thereof, and the flexible Schrader connector is engaged in the Schrader recess and has at least one Schrader retainer extending to an inner wall of the flexible Schrader connector; wherein the at least one Schrader retainer is connected with the at least one Schrader passing orifice, and an

extending length of the at least one Schrader retainer is greater than a size of the at least one Schrader passing orifice so that the at least one Schrader retainer extends out of the Schrader cutout, and the at least one Schrader retainer corresponds to the push needle.

6. The removal tool as claimed in claim 5, wherein the part of the at least one Schrader retainer has a Schrader arcuate groove extending therefrom.
